DSR Automation — Implementation Guide

Power Apps Formulas + Power BI DAX | Step-by-Step Reference

For use with: Microsoft Power Apps | SharePoint Excel | Power BI Desktop

How to Use This Document

This guide gives you every formula and DAX measure you need to build the DSR automation system. Follow the sections in order — Power Apps first, then Power BI.

- **Step 1:** Section 1: Excel table schema — set this up on SharePoint first
- **Step 2:** Section 2: Power Apps — all screens, controls, and Power Fx formulas
- **Step 3:** Section 3: Power BI — connect to SharePoint, all DAX measures, and report layout
- **Step 4:** Section 4: Power Automate — flow steps to connect everything

> **TIP:** All formulas are copy-paste ready. Replace placeholder names like [YourSharePointURL] with your actual values.

Section 1 — Excel Table Schema on SharePoint

Create one Excel file on your SharePoint shared drive named DSR_Data.xlsx. Inside it, create three tables using Insert > Table. Name each table exactly as shown.

1.1 Table: DSR_Master

One row per daily report submission. Name this table exactly DSR_Master (Table Design > Table Name).

Column Name	Type	Notes
DSR_ID	Number	Auto-increment. Formula: =ROW()-1
ProjectName	Text	From Power Apps dropdown
ReportDate	Date	Format: YYYY-MM-DD
Sprint	Text	e.g. Sprint 14
ReportedBy	Text	User display name
Period	Text	e.g. 21 Apr - 25 Apr
RAGStatus	Text	Values: Green / Amber / Red
PlannedTasks	Number	
DoneTasks	Number	
InProgressTasks	Number	
BlockedTasks	Number	
PctComplete	Number	Store as integer, e.g. 67
Highlights	Text	Multi-line text, pipe-separated
Accomplishments	Text	Multi-line text
Risks	Text	Multi-line text
Mitigation	Text	Multi-line text
NextSteps	Text	Plan for tomorrow
Dependencies	Text	
CriticalDefects	Number	
HighDefects	Number	
MediumDefects	Number	
LowDefects	Number	
TotalOpenDefects	Number	
ClosedTodayDefects	Number	
DefectNotes	Text	Free text notes
SubmittedAt	DateTime	Auto-set by Power Apps: Now()

1.2 Table: Jira_Details

One row per Jira ticket per DSR submission. Name this table Jira_Details.

Column Name	Type	Notes
JiraRow_ID	Number	Auto-increment
DSR_ID	Number	Foreign key — matches DSR_Master.DSR_ID
ReportDate	Date	Duplicate for easy filtering in Power BI
ProjectName	Text	Duplicate for easy filtering
TicketID	Text	e.g. PROJ-101
Summary	Text	Ticket title

Assignee	Text	
Status	Text	Done / In Progress / Blocked / Review / Open
Priority	Text	Critical / High / Medium / Low
Sprint	Text	

IMPORTANT: Keep the table names exact — DSR_Master and Jira_Details — because Power Apps and Power BI reference them by name.

Section 2 — Power Apps: All Screens & Formulas

2.1 Initial Setup

- Open Power Apps (make.powerapps.com) > Create > Blank canvas app > Tablet layout
- Add data source: Data > Add data > search "SharePoint" > paste your SharePoint site URL > select DSR_Data.xlsx > select both tables DSR_Master and Jira_Details
- Add a global variable on App.OnStart (select App in tree view, go to OnStart property)

App.OnStart formula

TIP: Put this in App > OnStart property. It runs when the app loads and sets up defaults.

```
Set(gCurrentUser, User().FullName);
Set(gToday, Today());
ClearCollect(colJira, Blank());
Set(gDSR, {
    ProjectName: "",
    ReportDate: Today(),
    Sprint: "",
    Period: "",
    RAGStatus: "Green",
    PlannedTasks: 0,
    DoneTasks: 0,
    InProgressTasks: 0,
    BlockedTasks: 0,
    Highlights: "",
    Accomplishments: "",
    Risks: "",
    Mitigation: "",
    NextSteps: "",
    Dependencies: "",
    CriticalDefects: 0,
    HighDefects: 0,
    MediumDefects: 0,
    LowDefects: 0,
    ClosedTodayDefects: 0,
    DefectNotes: ""
});
```

2.2 Screen 1 — Project Info

Add these controls on Screen 1. Set each property as shown.

Control Name	Control Type	Property	Formula / Value
txtProject Name	Text Input	Default	gCurrentUser
txtProject Name	Text Input	HintText	"Enter project name"
dtReportD	Date	Default	Today()

ate	Picker	Date	
txtSprint	Text Input	Default	""
txtPeriod	Text Input	HintText	"e.g. 21 Apr - 25 Apr"
ddRAG	Dropdown	Items	["Green","Amber","Red"]
ddRAG	Dropdown	Default	"Green"
lblRAGColor	Label	Color	If(ddRAG.Selected.Value="Green",RGBA(56,118,29,1),ddRAG.Selected.Value="Amber",RGBA(191,144,0,1),RGBA(192,0,0,1))
btnNext1	Button	Text	"Next >"
btnNext1	Button	OnSelect	See formula below

btnNext1.OnSelect — Screen 1 to Screen 2

TIP: This validates required fields, saves to gDSR, then navigates.

```
If(
    IsBlank(txtProjectName.Text),
    Notify("Project name is required", NotificationType.Error),
    IsBlank(txtSprint.Text),
    Notify("Sprint is required", NotificationType.Error),
    // All valid - save and navigate
    Set(gDSR, Patch(gDSR, {
        ProjectName:  txtProjectName.Text,
        ReportDate:   dtReportDate.SelectedDate,
        Sprint:       txtSprint.Text,
        Period:       txtPeriod.Text,
        RAGStatus:    ddRAG.Selected.Value
    }));
    Navigate(Screen2, ScreenTransition.Fade)
);
```

2.3 Screen 2 — Progress & Highlights

Control Name	Control Type	Property	Formula / Value
txtPlanned	Text Input	Default	"0"
txtPlanned	Text Input	Format	TextFormat.Number
txtDone	Text Input	Default	"0"
txtInProg	Text	Def	"0"

ress	Input	ault	
txtBlocked	Text Input	Default	"0"
lblPctComplete	Label	Text	Text(Round(If(Value(txtPlanned.Text)=0,0,Value(txtDone.Text)/Value(txtPlanned.Text)*100),0)) & "%"
lblPctComplete	Label	Color	If(Value(txtPlanned.Text)=0,RGBA(0,0,0,1),Value(txtDone.Text)/Value(txtPlanned.Text)>=0.8,RGBA(56,118,29,1),Value(txtDone.Text)/Value(txtPlanned.Text)>=0.5,RGBA(191,144,0,1),RGBA(192,0,0,1))
rectProgress	Rectangle	Width	If(Value(txtPlanned.Text)=0,0,400 * Value(txtDone.Text) / Value(txtPlanned.Text))
rectProgress	Rectangle	Fill	RGBA(56,118,29,1)
txtHighlights	Text Input	Mode	TextMode.MultiLine
txtHighlights	Text Input	HintText	"Enter one highlight per line"
txtAccomplishments	Text Input	Mode	TextMode.MultiLine

btnNext2.OnSelect — Screen 2 to Screen 3

```
Set(gDSR, Patch(gDSR, {
    PlannedTasks:    Value(txtPlanned.Text),
    DoneTasks:       Value(txtDone.Text),
    InProgressTasks: Value(txtInProgress.Text),
    BlockedTasks:    Value(txtBlocked.Text),
    PctComplete:     Round(If(Value(txtPlanned.Text)=0,0,
                     Value(txtDone.Text)/Value(txtPlanned.Text)*100),0),
    Highlights:      txtHighlights.Text,
    Accomplishments: txtAccomplishments.Text
}));
Navigate(Screen3, ScreenTransition.Fade);
```

2.4 Screen 3 — Defect Summary

Control Name	Control Type	Property	Formula / Value
txtCritical	Text Input	Default	"0"
txtHigh	Text Input	Default	"0"
txtMedium	Text Input	Default	"0"

txtLow	Text Input	Default	"0"
lblTotalOpen	Label	Text	Text(Value(txtCritical.Text)+Value(txtHigh.Text)+Value(txtMedium.Text)+Value(txtLow.Text))
txtClosedToday	Text Input	Default	"0"
txtDefectNotes	Text Input	Mode	TextMode.MultiLine
lblCritColor	Label	Color	If(Value(txtCritical.Text)>0,RGBA(192,0,0,1),RGBA(0,0,0,1))

btnNext3.OnSelect — Screen 3 to Screen 4

```
Set(gDSR, Patch(gDSR, {
    CriticalDefects:     Value(txtCritical.Text),
    HighDefects:         Value(txtHigh.Text),
    MediumDefects:       Value(txtMedium.Text),
    LowDefects:          Value(txtLow.Text),
    TotalOpenDefects:    Value(txtCritical.Text)+Value(txtHigh.Text)
                         +Value(txtMedium.Text)+Value(txtLow.Text),
    ClosedTodayDefects:  Value(txtClosedToday.Text),
    DefectNotes:         txtDefectNotes.Text
}));
Navigate(Screen4, ScreenTransition.Fade);
```

2.5 Screen 4 — Path to Green

Control Name	Control Type	Property	Formula / Value
txtRisks	Text Input	Mode	TextMode.MultiLine
txtRisks	Text Input	HintText	"Enter each risk on a new line"
txtMitigation	Text Input	Mode	TextMode.MultiLine
txtNextSteps	Text Input	Mode	TextMode.MultiLine
txtNextSteps	Text Input	Default	LookUp(DSR_Master, ProjectName=gDSR.ProjectName && ReportDate=Today()-1, NextSteps)
txtDependencies	Text Input	Mode	TextMode.MultiLine

TIP: The NextSteps Default formula auto-populates from yesterday's report for the same project — saves time for recurring items.

btnNext4.OnSelect — Screen 4 to Screen 5

```
Set(gDSR, Patch(gDSR, {
    Risks:         txtRisks.Text,
    Mitigation:    txtMitigation.Text,
    NextSteps:     txtNextSteps.Text,
    Dependencies: txtDependencies.Text
}));
Navigate(Screen5, ScreenTransition.Fade);
```

2.6 Screen 5 — Jira Task Table

This screen uses a Gallery control to show the Jira rows, and a separate overlay screen (Screen5b) as a modal form to add new rows.

Control Name	Control Type	Property	Formula / Value
galJira	Gallery	Items	colJira
galJira	Gallery	Layout	Layout.Table
lblTicketID	Label (in gallery)	Text	ThisItem.TicketID
lblSummary	Label (in gallery)	Text	ThisItem.Summary
lblAssignee	Label (in gallery)	Text	ThisItem.Assignee
lblStatus	Label (in gallery)	Text	ThisItem.Status
lblStatus	Label (in gallery)	Color	If(ThisItem.Status="Done",RGBA(56,118,29,1),ThisItem.Status="Blocked",RGBA(192,0,0,1),ThisItem.Status="In Progress",RGBA(31,73,125,1),RGBA(89,89,89,1))
lblPriority	Label (in gallery)	Text	ThisItem.Priority
btnDeleteRow	Button (in gallery)	Text	"X"
btnDeleteRow	Button (in gallery)	OnSelect	Remove(colJira, ThisItem)
btnAddRow	Button	Text	"+ Add Jira Row"
btnAddRow	Button	OnSelect	Navigate(Screen5b, ScreenTransition.None)

Screen5b (Add Jira Row modal) — btnSaveJira.OnSelect

```
Collect(colJira, {
    TicketID: txtJiraID.Text,
    Summary:  txtJiraSummary.Text,
    Assignee: txtJiraAssignee.Text,
```

```
    Status:   ddJiraStatus.Selected.Value,
    Priority: ddJiraPriority.Selected.Value
});
Back();
```

Dropdown Items for Status and Priority

```
// ddJiraStatus.Items:
["Done", "In Progress", "Blocked", "Review", "Open"]

// ddJiraPriority.Items:
["Critical", "High", "Medium", "Low"]
```

2.7 Screen 6 — Review & Submit

This is the final screen. It shows a summary and then writes all data to SharePoint on submit.

Control Name	Control Type	Property	Formula / Value
lblSummaryProject	Label	Text	gDSR.ProjectName
lblSummaryDate	Label	Text	Text(gDSR.ReportDate, "dd mmm yyyy")
lblSummaryProgress	Label	Text	gDSR.PctComplete & "% — " & gDSR.DoneTasks & "/" & gDSR.PlannedTasks & " tasks"
lblSummaryDefects	Label	Text	gDSR.TotalOpenDefects & " open (" & gDSR.CriticalDefects & " critical)"
lblSummaryRAG	Label	Text	gDSR.RAGStatus
lblSummaryRAG	Label	Color	If(gDSR.RAGStatus="Green",RGBA(56,118,29,1),gDSR.RAGStatus="Amber",RGBA(191,144,0,1),RGBA(192,0,0,1))
btnSubmit	Button	Text	"Submit DSR Report"
btnSubmit	Button	OnSelect	See formula below

btnSubmit.OnSelect — The Main Submit Formula

IMPORTANT: This is the most important formula. It writes to DSR_Master, then loops through all Jira rows, then triggers Power Automate.

```
// Step 1 — Write main DSR row to SharePoint
Set(varNewRecord,
    Patch(DSR_Master, Defaults(DSR_Master), {
        ProjectName:          gDSR.ProjectName,
        ReportDate:           gDSR.ReportDate,
        Sprint:               gDSR.Sprint,
        ReportedBy:           gCurrentUser,
        Period:               gDSR.Period,
        RAGStatus:            gDSR.RAGStatus,
        PlannedTasks:         gDSR.PlannedTasks,
```

```
        DoneTasks:           gDSR.DoneTasks,
        InProgressTasks:     gDSR.InProgressTasks,
        BlockedTasks:        gDSR.BlockedTasks,
        PctComplete:         gDSR.PctComplete,
        Highlights:          gDSR.Highlights,
        Accomplishments:     gDSR.Accomplishments,
        Risks:               gDSR.Risks,
        Mitigation:          gDSR.Mitigation,
        NextSteps:           gDSR.NextSteps,
        Dependencies:        gDSR.Dependencies,
        CriticalDefects:     gDSR.CriticalDefects,
        HighDefects:         gDSR.HighDefects,
        MediumDefects:       gDSR.MediumDefects,
        LowDefects:          gDSR.LowDefects,
        TotalOpenDefects:    gDSR.TotalOpenDefects,
        ClosedTodayDefects:  gDSR.ClosedTodayDefects,
        DefectNotes:         gDSR.DefectNotes,
        SubmittedAt:         Now()
    })
);

// Step 2 — Write each Jira row
ForAll(colJira,
    Patch(Jira_Details, Defaults(Jira_Details), {
        DSR_ID:      varNewRecord.DSR_ID,
        ReportDate:  gDSR.ReportDate,
        ProjectName: gDSR.ProjectName,
        TicketID:    TicketID,
        Summary:     Summary,
        Assignee:    Assignee,
        Status:      Status,
        Priority:    Priority,
        Sprint:      gDSR.Sprint
    })
);

// Step 3 — Trigger Power Automate flow
// (Replace YourFlowName with your actual flow name)
YourFlowName.Run(
    gDSR.ProjectName,
    Text(gDSR.ReportDate, "dd mmm yyyy"),
    gDSR.RAGStatus
);

// Step 4 — Show confirmation and reset
Notify("DSR submitted successfully!", NotificationType.Success);
Set(gDSR, Blank());
```

```
ClearCollect(colJira, Blank());
Navigate(Screen1, ScreenTransition.Fade);
```

Section 3 — Power BI: Connection & DAX Measures

3.1 Connect Power BI to SharePoint Excel

1. Open Power BI Desktop
2. Click Get Data > More > Online Services > SharePoint Online List
3. Alternatively: Get Data > Excel Workbook — paste the SharePoint URL of DSR_Data.xlsx
4. Select both tables: DSR_Master and Jira_Details
5. Click Transform Data to open Power Query
6. In Power Query: ensure ReportDate is Date type, all numeric columns are Whole Number type
7. Close & Apply

TIP: SharePoint URL format: https://yourcompany.sharepoint.com/sites/YourSite/Shared Documents/DSR_Data.xlsx

3.2 Create a Date Table (Required for time intelligence)

In Power BI, go to Modeling > New Table and paste this DAX:

```
DateTable =
ADDCOLUMNS(
    CALENDAR(DATE(2024,1,1), DATE(2026,12,31)),
    "Year",         YEAR([Date]),
    "Month",        FORMAT([Date], "MMM YYYY"),
    "MonthNum",     MONTH([Date]),
    "Week",         WEEKNUM([Date]),
    "DayName",      FORMAT([Date], "dddd"),
    "IsWeekday",    IF(WEEKDAY([Date],2) <= 5, TRUE, FALSE)
)
```

Then create a relationship: DateTable[Date] -> DSR_Master[ReportDate] (many-to-one).

3.3 DAX Measures — Create these in DSR_Master table

TIP: To create a measure: click DSR_Master in the Fields pane > New Measure. Paste each formula below.

Basic Counts

```
// Total planned tasks
Total Planned = SUM(DSR_Master[PlannedTasks])

// Total done tasks
Total Done = SUM(DSR_Master[DoneTasks])

// In progress
Total In Progress = SUM(DSR_Master[InProgressTasks])

// Blocked
Total Blocked = SUM(DSR_Master[BlockedTasks])

// Latest % complete (most recent report)
Latest Pct Complete =
```

```
CALCULATE(
    MAX(DSR_Master[PctComplete]),
    DSR_Master[ReportDate] = MAX(DSR_Master[ReportDate])
)
```

Defect Measures

```
// Total open defects (latest day)
Latest Open Defects =
CALCULATE(
    SUM(DSR_Master[TotalOpenDefects]),
    DSR_Master[ReportDate] = MAX(DSR_Master[ReportDate])
)

// Critical defects today
Critical Defects Today =
CALCULATE(
    SUM(DSR_Master[CriticalDefects]),
    DSR_Master[ReportDate] = MAX(DSR_Master[ReportDate])
)

// Defects closed today
Defects Closed Today =
CALCULATE(
    SUM(DSR_Master[ClosedTodayDefects]),
    DSR_Master[ReportDate] = MAX(DSR_Master[ReportDate])
)

// Defect severity breakdown for donut chart
Defect Critical = SUM(DSR_Master[CriticalDefects])
Defect High     = SUM(DSR_Master[HighDefects])
Defect Medium   = SUM(DSR_Master[MediumDefects])
Defect Low      = SUM(DSR_Master[LowDefects])
```

RAG Status Measures

```
// Latest RAG (for KPI card)
Latest RAG =
CALCULATE(
    FIRSTNONBLANK(DSR_Master[RAGStatus], 1),
    DSR_Master[ReportDate] = MAX(DSR_Master[ReportDate])
)

// Count of Green days in selected period
Green Days =
CALCULATE(
    COUNTROWS(DSR_Master),
    DSR_Master[RAGStatus] = "Green"
)
```

```
// Count of Red days
Red Days =
CALCULATE(
    COUNTROWS(DSR_Master),
    DSR_Master[RAGStatus] = "Red"
)
```

Trend & Comparison Measures

```
// % complete vs previous day
Pct Complete vs Yesterday =
VAR Today    = MAX(DSR_Master[ReportDate])
VAR Yesterday = Today - 1
VAR TodayPct =
    CALCULATE(MAX(DSR_Master[PctComplete]),
              DSR_Master[ReportDate] = Today)
VAR YestPct =
    CALCULATE(MAX(DSR_Master[PctComplete]),
              DSR_Master[ReportDate] = Yesterday)
RETURN TodayPct - YestPct

// Rolling 7-day defect average
Defect 7Day Avg =
CALCULATE(
    AVERAGE(DSR_Master[TotalOpenDefects]),
    DATESINPERIOD(DSR_Master[ReportDate], MAX(DSR_Master[ReportDate]), -7, DAY)
)

// Sprint velocity (done vs planned ratio)
Sprint Velocity % =
DIVIDE([Total Done], [Total Planned], 0) * 100
```

Jira Table Measures

```
// Count tickets by status (use in Jira table or bar chart)
Tickets Done =
CALCULATE(COUNTROWS(Jira_Details), Jira_Details[Status] = "Done")

Tickets Blocked =
CALCULATE(COUNTROWS(Jira_Details), Jira_Details[Status] = "Blocked")

Tickets In Progress =
CALCULATE(COUNTROWS(Jira_Details), Jira_Details[Status] = "In Progress")

// Completion % per assignee (use in assignee bar chart)
Assignee Completion % =
DIVIDE(
    CALCULATE(COUNTROWS(Jira_Details), Jira_Details[Status] = "Done"),
```

```
    COUNTROWS(Jira_Details),
    0
) * 100
```

3.4 Conditional Formatting for RAG

To color a KPI card or table cell based on RAG, use these DAX color measures:

```
// RAG Color (returns hex code)
RAG Color =
SWITCH(
    [Latest RAG],
    "Green", "#38761D",
    "Amber", "#BF9000",
    "Red",   "#C00000",
    "#595959"
)
```

Apply this: select your visual > Format > Conditional Formatting > choose "RAG Color" measure as field value.

3.5 Report Layout — Build Order

Order	Visual Type	Fields / Measure	Position
1	Slicers (4)	ProjectName, Sprint, ReportDate, RAGStatus	Top row
2	Card	Latest Pct Complete	KPI row left
3	Card	Latest Open Defects	KPI row
4	Card	Critical Defects Today	KPI row
5	Card	Total Blocked	KPI row
6	Card	Sprint Velocity %	KPI row right
7	Line Chart	X: ReportDate, Y: PctComplete, Y2: PlannedTasks	Middle left — burndown
8	Donut Chart	Values: Defect Critical/High/Medium/Low	Middle centre
9	Matrix / Table	Rows: ReportDate, Values: RAGStatus colored	Middle right — RAG history
10	Clustered Bar	Axis: Assignee, Value: Assignee Completion %	Lower left
11	Line Chart	X: ReportDate, Y: Total Open Defects	Lower right — trend
12	Table	TicketID, Summary, Assignee, Status, Priority, ReportDate	Bottom — Jira table
13	Text Box x3	Risks, Mitigation, NextSteps from card visual	Footer row

TIP: Enable auto-refresh: Publish the report to Power BI Service, then go to Dataset Settings > Scheduled Refresh > set to daily (e.g. 8:00 AM before standup).

Section 4 — Power Automate Flows

4.1 Flow 1 — Send Teams/Email Notification on DSR Submit

8. Go to make.powerautomate.com > New Flow > Instant cloud flow
9. Trigger: "PowerApps (V2)" — this is what Power Apps calls via YourFlowName.Run()
10. Add input: ProjectName (Text), ReportDate (Text), RAGStatus (Text)
11. Add action: "Post message in a chat or channel" (Microsoft Teams)
12. Set Channel to your team channel and compose the message:

```
DSR Submitted: @{triggerBody()['ProjectName']}
Date: @{triggerBody()['ReportDate']}
Status: @{triggerBody()['RAGStatus']}
View report: [paste your Power BI report URL here]
```

13. (Optional) Add action: "Send an email (V2)" for email digest
14. Save the flow. Go back to Power Apps and connect it via the Power Automate pane

4.2 Flow 2 — Daily Power BI Dataset Refresh (Scheduled)

15. New flow > Scheduled cloud flow > set to daily at 8:00 AM
16. Add action: "Refresh a dataset" (Power BI)
17. Select your Workspace and DSR dataset
18. Save — this ensures Power BI has fresh data before morning standup

IMPORTANT: You need a Power BI Pro license or Premium workspace to use scheduled refresh via Power Automate. If you only have free Power BI, refresh manually by clicking "Refresh now" in the dataset settings.

Section 5 — Common Issues & Fixes

Issue	Likely Cause	Fix
Patch() fails silently	Column name mismatch	Check exact column names in Excel table match exactly what you write in Power Fx
ForAll() does not write all rows	Delegation limit	Add ClearCollect to a local collection first; ForAll works on local collections without delegation limits
Power BI shows no data	Wrong data type in Power Query	In Power Query set ReportDate to Date type and all number columns to Whole Number
Relationship error in Power BI	DSR_ID not matching	In Power BI Model view, drag DSR_Master[DSR_ID] to Jira_Details[DSR_ID] to create the relationship
% Complete shows > 100%	Multiple rows per day	Use MAX(PctComplete) not SUM in your measure
Gallery shows blank	colJira is empty on load	Ensure App.OnStart has ClearCollect(colJira, Blank()) — this initialises the collection
Teams notification not firing	Flow not connected in Power Apps	In Power Apps: Add > Power Automate > Add your flow. Then reference it in the OnSelect formula
Slicer not filtering Jira table	No relationship between tables	Create relationship: DSR_Master[DSR_ID] one-to-many Jira_Details[DSR_ID] in Power BI Model view